

02 U.S. Case

Institutional democracy and group-centered power-sharing

Jessala A. Grijalva

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Purpose

This file applies the Power-Sharing Index (PSI) to the U.S. case after the measure has been constructed in `01_measurement.qmd`. It compares group-centered power-sharing with established V-Dem democracy indices, estimates a power-sharing deficit, and runs simple descriptive diagnostics to identify what belongs in the manuscript, appendix, or replication materials.

Setup

Loaded Derived PSI Data

Derived file	First year	Last year	Country-years
data/derived/usa_psi_final.rda	1789	2025	237

V-Dem Comparison Measures

Available U.S. coverage in the derived PSI dataset

Measure	Variable	Coverage
Electoral Democracy	v2x_polyarchy	1789-2025
Liberal Democracy	v2x_libdem	1789-2025
Participatory Democracy	v2x_partipdem	1789-2025
Deliberative Democracy	v2x_delibdem	1900-2025
Egalitarian Democracy	v2x_egaldem	1900-2025

Load derived PSI data

This file does not rebuild the PSI. It loads the final derived object created by `01_measurement.qmd`, which keeps the workflow modular and makes the measurement file the single source of truth.

Define comparison measures

The additive PSI is used as the main descriptive score because it is transparent and directly interpretable. The geometric and hybrid specifications remain available for robustness checks, and the institutional comparison is anchored in V-Dem's Electoral Democracy Index because it has full long-run coverage for the U.S. series.

External procedural comparator

Polity2 is carried forward as an optional external procedural-democracy comparator when it is available in the V-Dem file. It is not part of PSI construction. It is used only to check whether the power-sharing pattern also appears against a non-V-Dem democracy scale. If no usable Polity2 data are available, all Polity outputs are omitted cleanly.

External Procedural Comparator

Polity2 coverage in the U.S. case

Variable	Source	Coverage	Usable country-years	Scale
polity2_01	e_polity2	1789-2018	230	Polity2 rescaled from -10 to 10 onto 0 to 1

Descriptive trajectories

The first descriptive comparison places PSI beside Electoral Democracy on a common within-case scale. Electoral Democracy is expressed as its position within the U.S. series' observed historical range, while PSI is the additive power-sharing score constructed in the measurement file. This is the cleanest diagnostic for the paper's central contrast: institutional democratic development and group-centered power-sharing are related, but they are not identical.

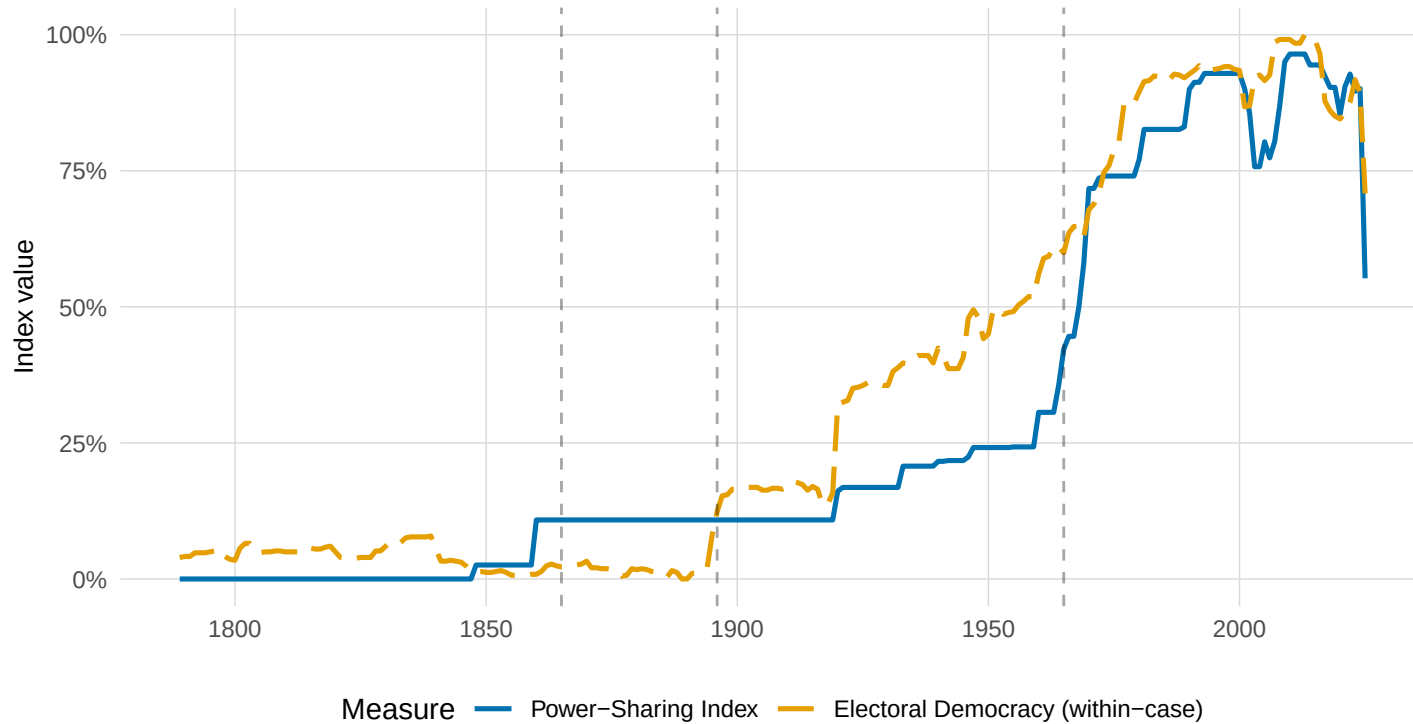
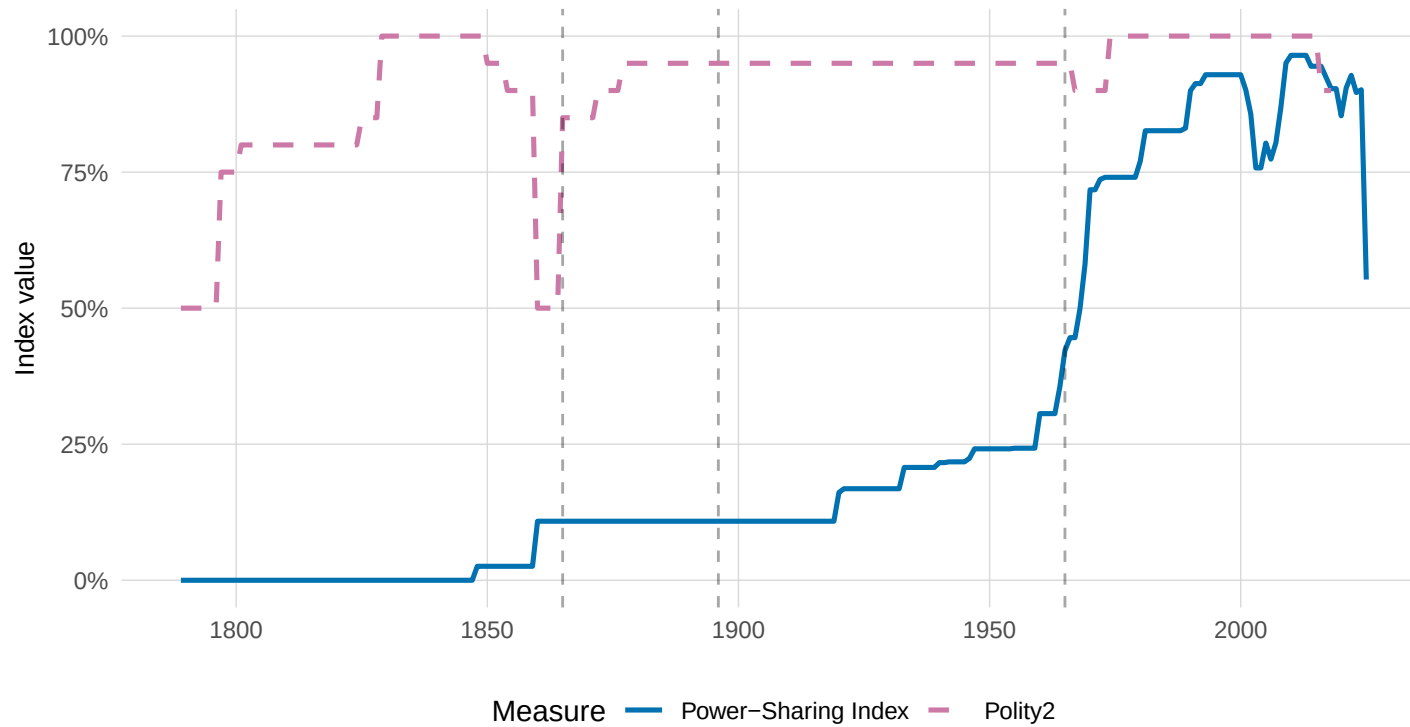


Figure 1: PSI and Electoral Democracy (within-case scale) in the U.S. case.

The Polity comparison is a robustness check against an external procedural-democracy scale. If PSI diverges from Polity2 as well as from V-Dem democracy indices, the group-centered power-sharing pattern is less dependent on V-Dem's internal democracy measurement architecture.



Broader V-Dem comparison from 1900

The broader five-index comparison begins in 1900 because Egalitarian Democracy and Deliberative Democracy do not provide coverage before 1900 in the U.S. series. Restricting this panel to 1900 onward keeps every plotted index on its actually observed support rather than implying pre-1900 values that V-Dem does not supply. The full-period institutional comparison remains the Electoral Democracy contrast above, which covers 1789 onward.

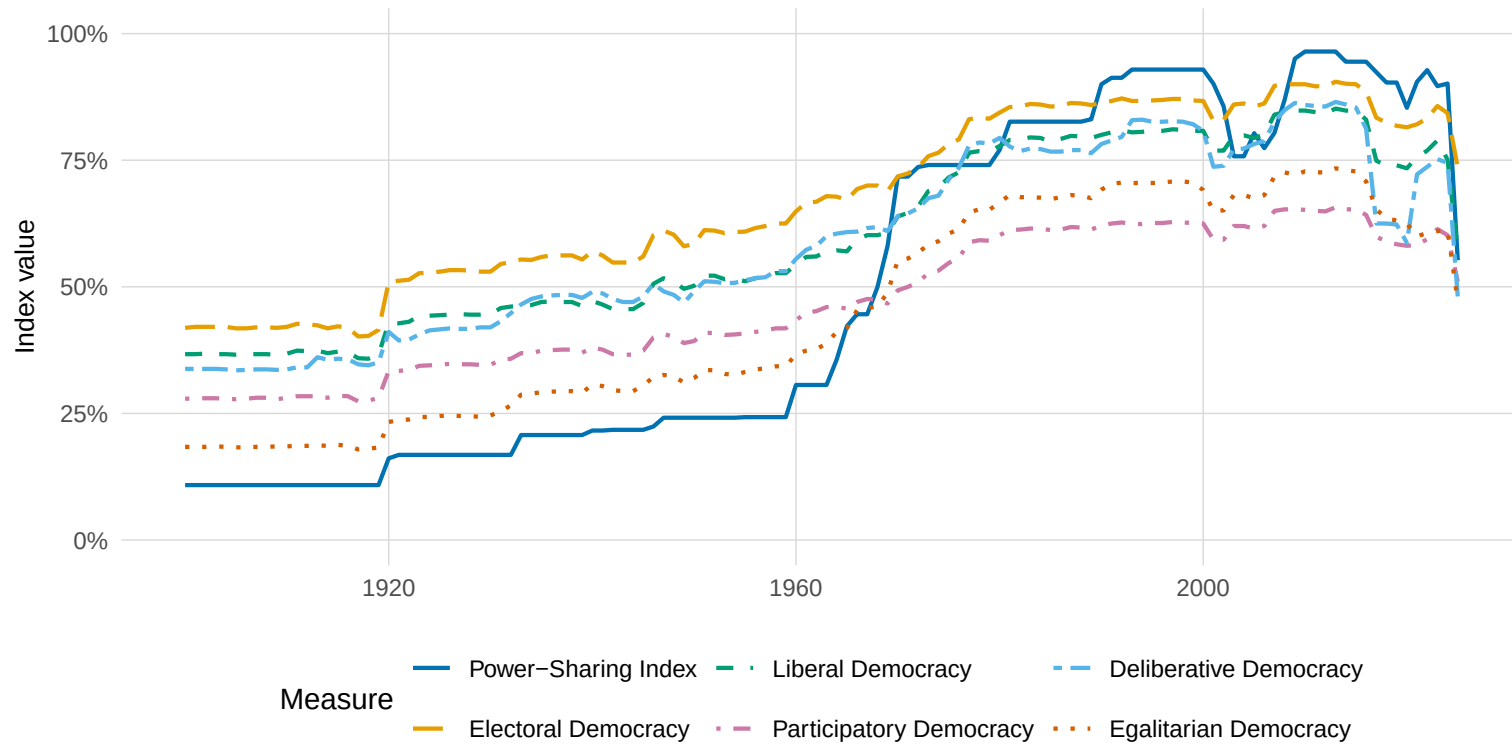


Figure 2: PSI compared with V-Dem democracy indices, 1900 onward. Egalitarian and Deliberative Democracy are unavailable before 1900.

Power-sharing deficit

The power-sharing deficit is defined symmetrically within the polity's own history: Electoral Democracy is expressed as its position within the U.S. series' observed range and PSI is the additive power-sharing score constructed in the measurement file. Positive values indicate years in which institutional democratic development exceeds group-centered power-sharing. The asymmetric definition using globally scaled Electoral Democracy is retained as a robustness specification. If V-Dem uncertainty columns are available, the figure includes a conservative uncertainty ribbon.

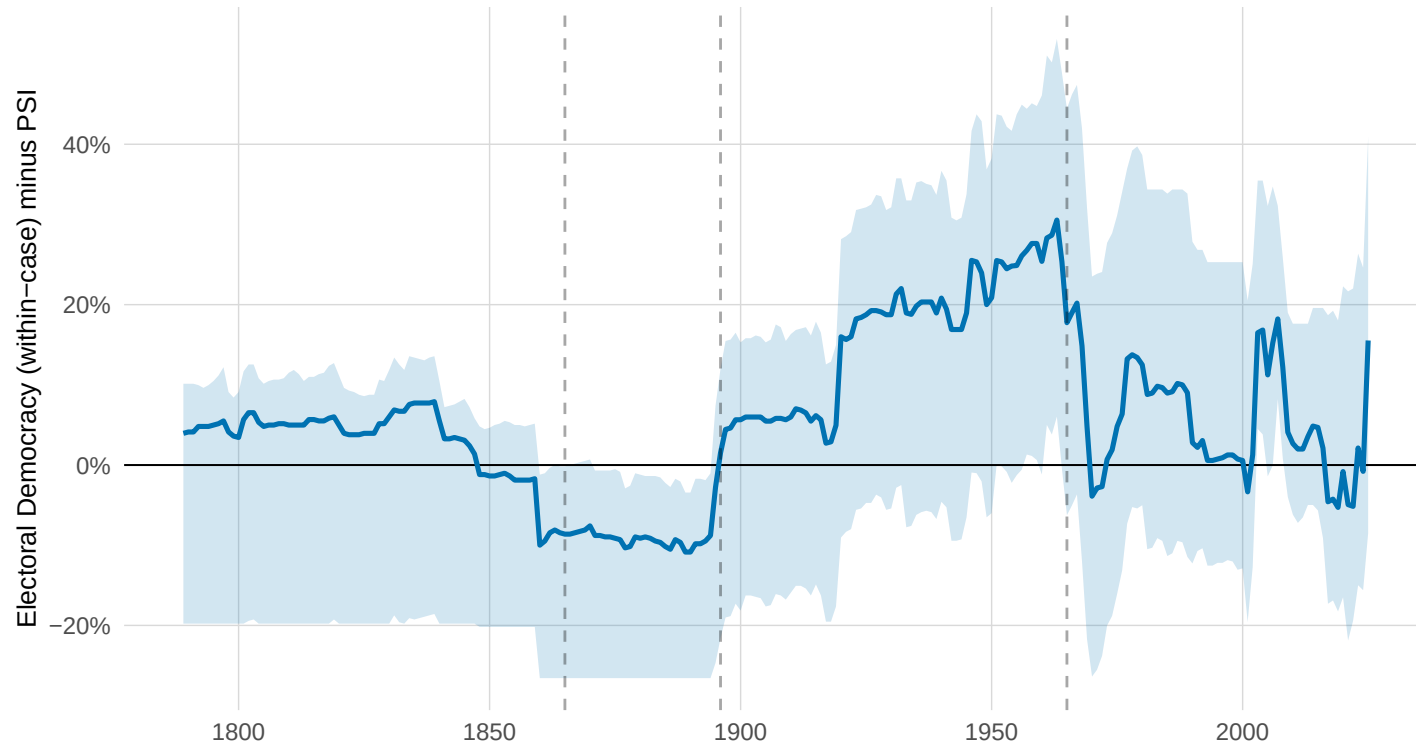


Figure 3: Power-sharing deficit using Electoral Democracy as the institutional baseline.

External-comparator deficit

The Polity-based deficit uses the same logic as the Electoral Democracy deficit, but substitutes the rescaled Polity2 score as the procedural-democracy baseline. This is a robustness check rather than the main empirical estimand.

Era-Level Summary

PSI, Electoral Democracy, and the power-sharing deficit

era	N	PSI	Electoral (global)	Electoral (within-case)	Deficit	Max deficit
Foundational Exclusion (1789-1865)	77	0.012	0.347	0.042	0.029	0.079
First Opening for Multiracial Democracy (1866-1896)	31	0.108	0.335	0.021	-0.087	0.015
Partial Inclusion and Retrenchment (1897-1964)	68	0.182	0.523	0.344	0.162	0.305
Second Opening for Multiracial Democracy (1965-present)	61	0.821	0.833	0.876	0.056	0.202

Polity-Based Power-Sharing Deficit

External procedural baseline

era	N	PSI	Polity2	Polity deficit
Foundational Exclusion (1789-1865)	77.000	0.012	0.820	0.808
First Opening for Multiracial Democracy (1866-1896)	31.000	0.108	0.923	0.814
Partial Inclusion and Retrenchment (1897-1964)	68.000	0.182	0.950	0.768
Second Opening for Multiracial Democracy (1965-present)	54.000	0.817	0.980	0.163

Component trajectories

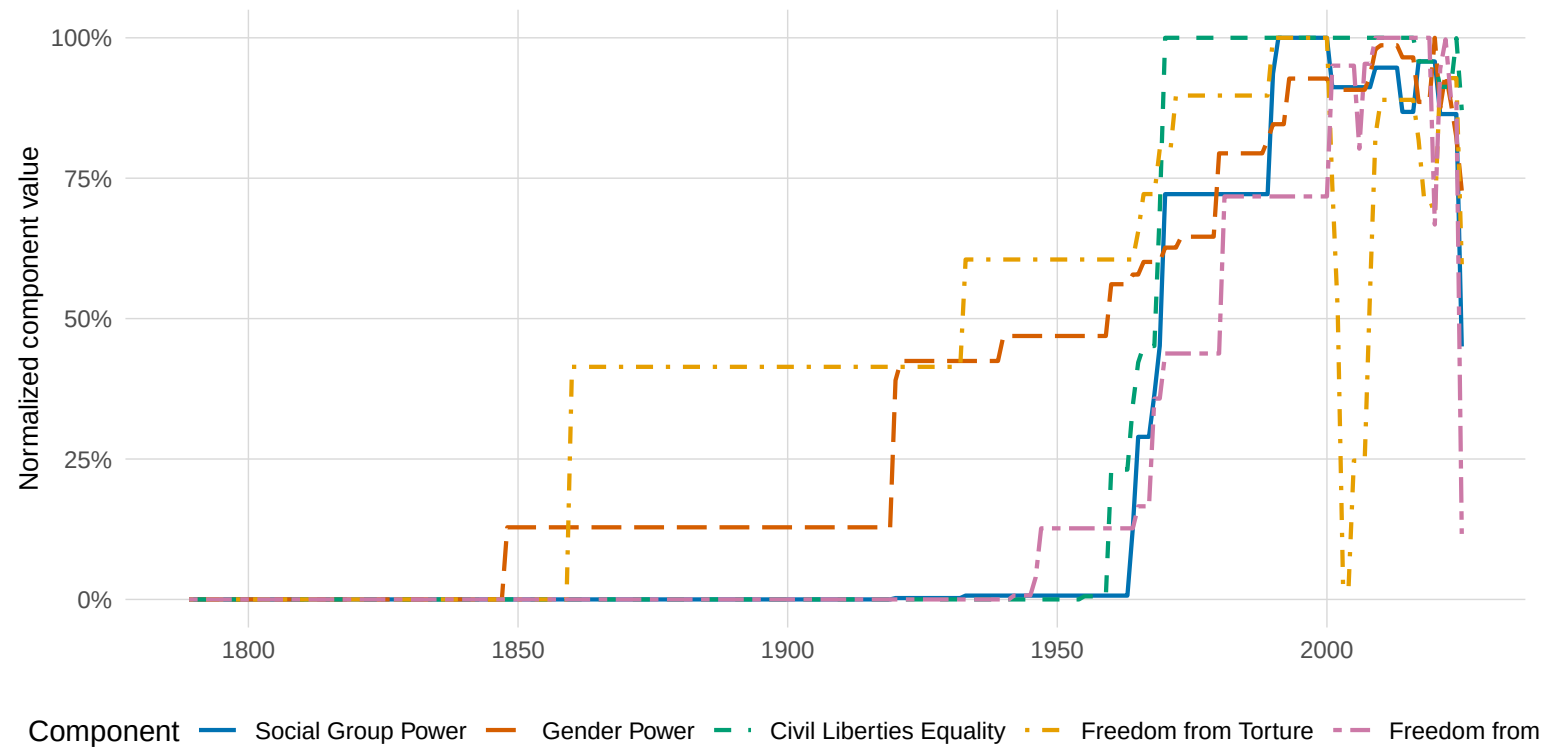


Figure 4: Normalized PSI component trajectories.

Robustness across PSI specifications

This check asks whether the power-sharing deficit depends on the additive PSI specification or the within-case scaling decision. If the deficit remains visible across additive, geometric, gated, and hybrid PSI variants, and under the asymmetric global Electoral Democracy definition, the substantive pattern is vulnerable neither to aggregation choice nor to the scaling decision.

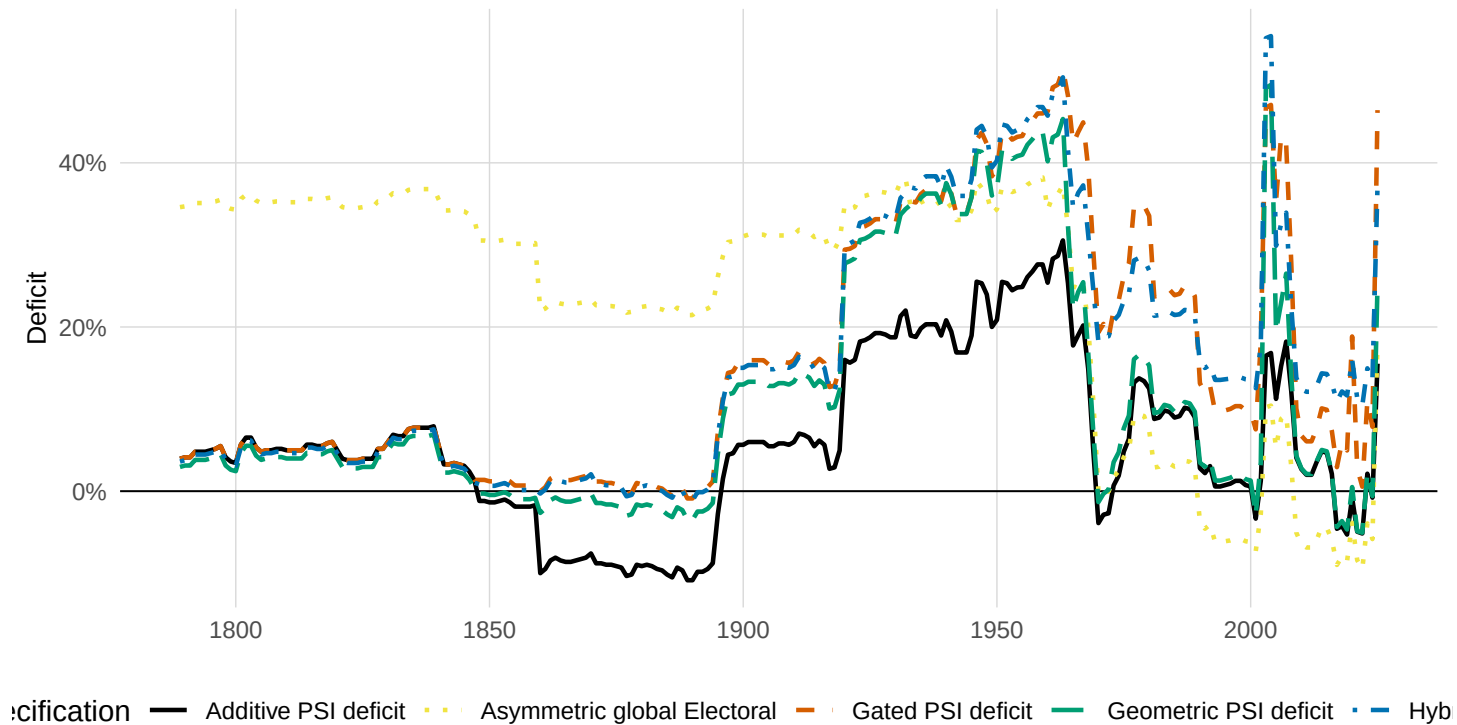


Figure 5: Power-sharing deficit across PSI specifications.

Temporal-ordering diagnostics

These diagnostics ask whether changes in group-centered power-sharing and institutional democracy move together or unevenly over time. They do not estimate causality; they are included to assess whether the observed trajectories raise a serious temporal-ordering question for future work.

Deficit Summary Across PSI Specifications

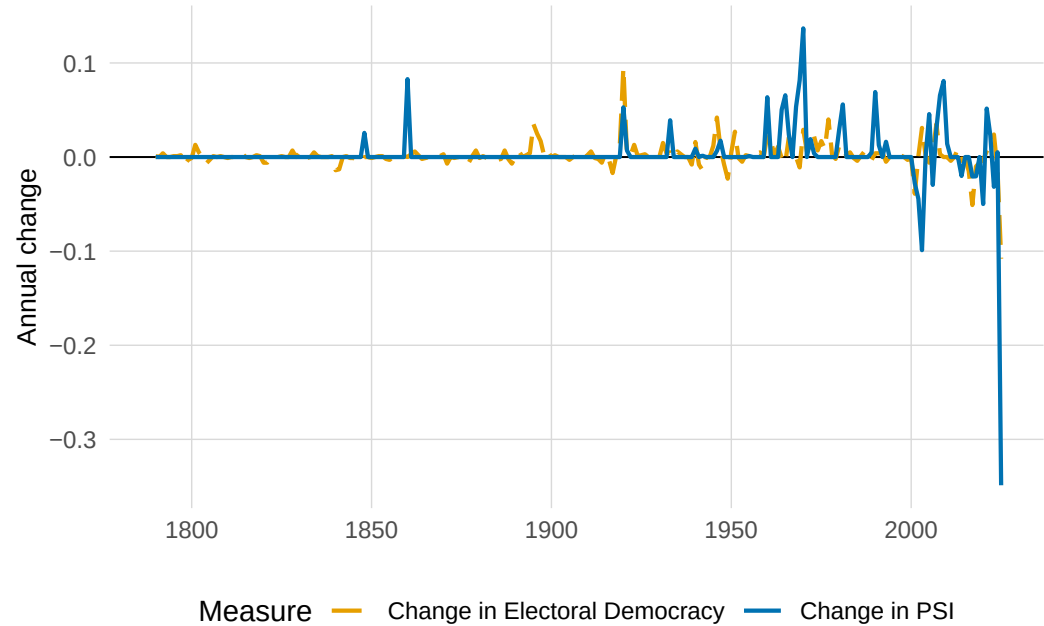
Symmetric within-case deficit; asymmetric definition included as scaling robustness

Specification	Mean	Median	Max
Additive PSI deficit	0.059	0.050	0.305
Asymmetric global Electoral	0.240	0.311	0.382
Gated PSI deficit	0.157	0.079	0.514
Geometric PSI deficit	0.111	0.045	0.495
Hybrid PSI deficit	0.156	0.125	0.554

Temporal-Ordering Diagnostic

Correlations between changes in PSI and changes in Electoral Democracy

Relationship	Correlation
Same-year Δ PSI and Δ Electoral Democracy	0.509
Δ PSI lagged 1 year and Δ Electoral Democracy	0.057
Δ PSI lagged 5 years and Δ Electoral Democracy	0.131
Δ PSI lagged 10 years and Δ Electoral Democracy	0.025



Candidate Outputs for Manuscript and Appendix

Output	Type	Purpose	Destination
PSI vs Electoral Democracy	Figure	Core institutional/group comparison	Manuscript
Power-sharing deficit	Figure	Main empirical payoff	Manuscript
Era summary	Table	Compact historical summary	Manuscript
PSI vs V-Dem indices	Figure	Comparison with established measures	Manuscript or appendix
Component trajectories	Figure	Composite transparency	Appendix
Deficit robustness	Figure/Table	Aggregation/gated sensitivity	Appendix
Polity comparison	Figure/Table	External procedural comparator if available	Appendix
Temporal diagnostics	Table/Figure	Raises temporal-ordering question	Appendix/discussion

Summary of U.S. Case Evidence

Finding	Evidence
PSI and Electoral Democracy are not identical	Trajectory comparison shows periods of divergence between PSI and Electoral Democracy
The power-sharing deficit is historically concentrated	Era summaries show the deficit is largest before full civil-rights incorporation
The second opening marks a large increase in group-centered power-sharing	PSI rises sharply in the post-1965 era across specifications
The composite is transparent to component inspection	Component trajectories show the histories of power, civil standing, and coercive restraint separately
The pattern is visible against an external comparator	Polity2 comparison provides a non-V-Dem procedural baseline
Temporal ordering warrants further study	Change-score diagnostics allow the question to be raised without making causal claims

Candidate manuscript and appendix outputs

This table is a decision aid. It separates analyses that appear central to the manuscript from analyses that are better suited for the appendix or retained as replication checks.

Summary of U.S. case evidence

Export U.S. case results

The modeling file saves one downstream object containing the main transformed data and summary tables. It does not export publication figures; those will be generated selectively when the manuscript or appendix requires them.

Saved Output

Output	Contents
data/derived/usa_case_results.rda	U.S. case results

Manuscript figure

This section assembles the main manuscript figure directly from the analysis objects created above. The figure is displayed in this PDF and exported once, deliberately, as `figures/GrijalvaFig1.pdf`.

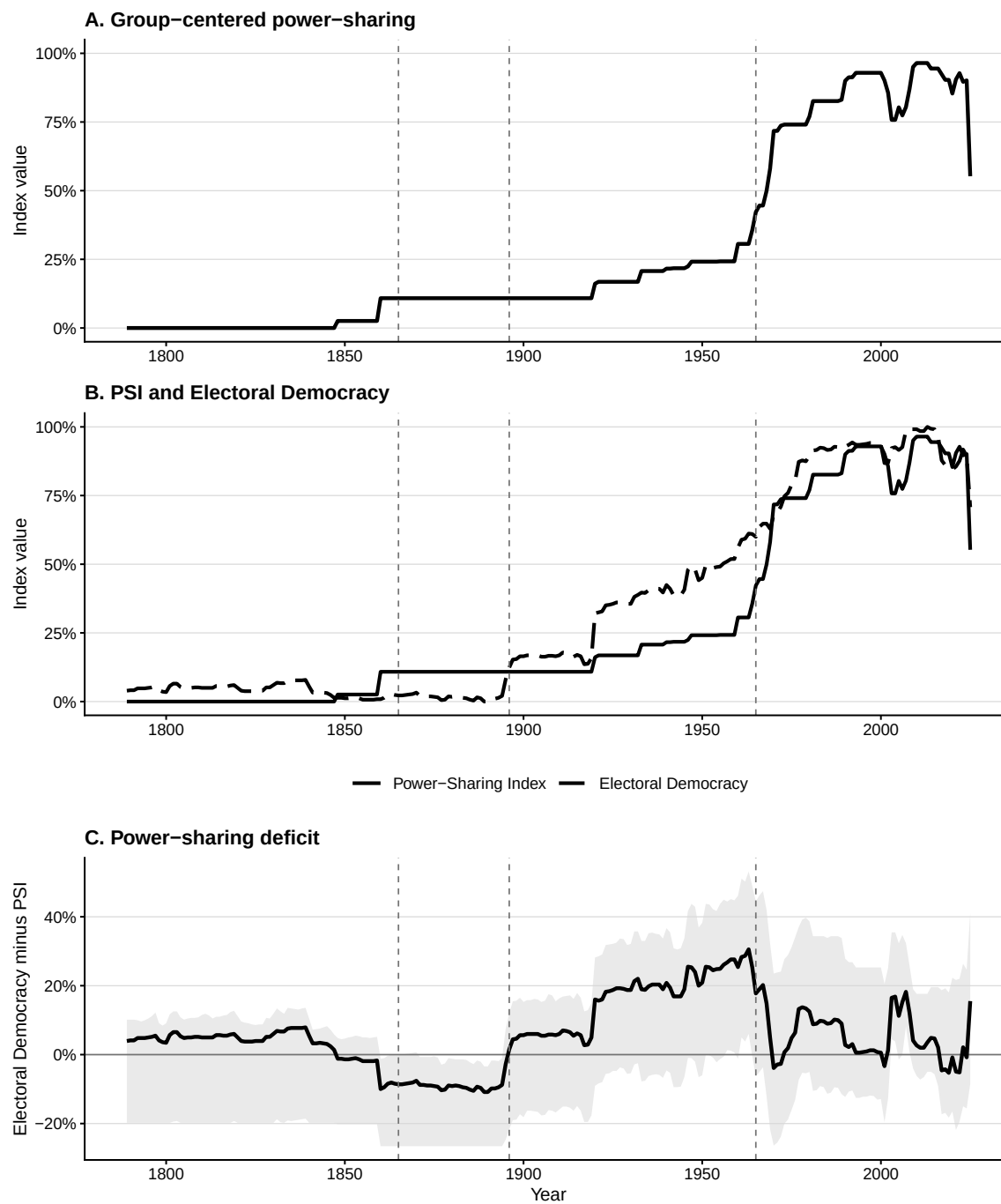


Figure 6: Figure 1. Group-centered power-sharing and the power-sharing deficit in the United States.

Exported Manuscript Figure

Output	Use
figures/GrijalvaFig1.pdf	Vector production file for manuscript submission

Figure 1. Group-centered power-sharing and the power-sharing deficit in the United States, 1789–2025. Panel A plots the additive Power-Sharing Index. Panel B compares PSI with V-Dem Electoral Democracy, rescaled within the U.S. historical series. Panel C plots the power-sharing deficit, defined as Electoral Democracy minus PSI. Higher PSI values indicate greater group-centered democratic power-sharing. Positive deficit values indicate years in which institutional democracy exceeds group-centered power-sharing.

Session info

R version 4.5.3 (2026-03-11)

Platform: aarch64-apple-darwin20

Running under: macOS Tahoe 26.5.2

Matrix products: default

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LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1

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other attached packages:

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loaded via a namespace (and not attached):

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